Robot-Assisted Autism Therapy for Children: Targeted Behaviors, Challenges, and Individual Differences

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ABSTRACT

A significant amount of studies over the past two decades have demonstrated that children with autism spectrum disorder (ASD) often are fascinated with robots. Robot-Assisted Autism Therapy (RAAT) is a therapeutic approach that utilizes robots as interactive tools to enhance various aspects of intervention and support for individuals with ASD. Despite the growing popularity of therapy with Socially Assistive Robots (SARs), RAAT has its challenges and the research about it is an ongoing process. This paper is a systematic literature review of scientific articles on RAAT, intending to identify its benefits, challenges, and research gaps. The main findings are: 1) Robots can be a valuable addition to traditional autism therapy, as they enhance specific behaviors during interventions. 2) The individual differences among children with autism have a significant impact and necessitate further investigation. 3) The effectiveness of using robots in real-world scenarios for long-term interventions requires more attention from the research community.

CCS CONCEPTS

• **Human-centered computing → Human computer interaction (HCI)**.

KEYWORDS

HRI, mental health, ASD, children, RAAT, SAR

1 INTRODUCTION

Imagine that you or someone you know has to face everyday challenges such as social communication, learning, and cognitive difficulties. These are just some of the most common symptoms for people with autism disorder that make their integration into the world difficult. Autism Spectrum Disorder (ASD) is a lifelong neurological and developmental disorder that affects individuals from a young age [16]. It is characterized by repetitive patterns of behaviors and impairments in cognition, motor, sensory, learning, and social skills [3, 16]. Traditional ASD therapy focuses on improving these skills through personalized sessions with one child at a time. One of the most used treatment principles is Applied Behavior Analysis (ABA). ABA enables prosocial and emotional behaviors through positive game-based reinforcement activities [4, 11, 16]. Improving therapy for children with autism (CWA) is an important topic since it can significantly elevate their life experiences. Being a spectrum disorder, autism can appear in different forms depending on the individual: low-functioning autism (LFA) and high-functioning autism (HFA) [11, 16]. Children with HFA have better cognitive and language abilities than children who are closer to LFA [11]. Moreover, autism usually coexists with other mental health conditions, more commonly with Attention Deficit and Hyperactivity Disorder (ADHD) [16]. Furthermore, research on ASD emphasizes the important influence of sensory preferences (e.g. overreliance on proprioceptive, hyporeactivity to visual cues) in ASD therapy [3]. Proprioception involves sensing body position and movement [3]. Overreliance on a specific sensory input suggests excessive dependence on it for environmental interpretation, while hyporeactivity indicates reduced responsiveness to specific sensory stimuli[3]. CWA display a broad spectrum of strengths and specific individual preferences [16]. Therefore, the heterogeneous form of autism plays a critical role in therapy since it can influence the therapy intervention outcomes [11, 16].

One of the newest additions to ASD therapy in the last century is using Socially Assistive Robots robots (SARs) that can offer mental health support through social interaction [5]. Robot-Assisted Autism Therapy (RAAT) is a specific area of research within Human-Robot Interaction (HRI) that focuses on using robots to support traditional therapy for individuals with autism. Research shows that children with autism (CWA) have a strong interest in robots [1, 9, 11] thanks to their interactive nature and repetitive behavior [12]. Moreover, CWA tend to imitate and be more engaged with a robot than with a human [3, 8]. As a result, social robots have a great potential to be effective therapeutic tools for CWA [2, 8, 10].

To address the use of robots in ASD therapy, I identify the benefits and challenges, as well as gaps in RAAT mentioned in the reviewed papers. This literature review addresses the following research questions:

(1) **RQ1: How can RAAT be a valuable addition to ASD therapy?**
(2) **RQ2: What are the challenges and gaps in RAAT?**

In this paper, I explore the current state of knowledge in HRI research on ASD therapy. A better understanding of RAAT is crucial not only for contributing to HRI research but also for enhancing the well-being of children with autism.

2 RESULTS

The first subsection illustrates the types of robots used in the studies and the session settings. The second subsection provides an overview of the targeted skills in interventions as identified in the reviewed literature. Finally, the third subsection analyzes the outcomes of the RAAT interventions.



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2.1 Robots and Setup

The reviewed experiments took place in a room without any other distractions. In some studies, the therapist was also present in the room with the child (e.g. Sandygulova et al. [11]), in others only the robot was present in the room with the child (e.g. Feil-Seifer et al. [6]). In the studies where the therapist was also present, the clinician customized the type of activities based on the children's reactions. In other studies, the therapist wasn't present in the room and the robot reacted autonomously to the child's positive or negative responses (e.g. Feil-Seifer et al. [6]). The researchers from the reviewed articles analyzed the RAAT interactions by recording the sessions. All studies used an autonomous robot SAR. Table 1 shows the used robots and how often they were used in the reviewed articles. All used robots had a humanoid design except an autonomous robotic dog AIBO used in Stanton et al. [13]. It's important to mention that the robot Blue used in Begum et al. [1] is very similar to the robot NAO and is fabricated by the same company.

2.2 Targeted Behaviors

To respond to research question RQ1, I analyzed the types of targeted behaviors in the interventions. My scope was to identify the skills targeted in traditional ASD therapy that can be carried out in RAAT and see if the reviewed studies consider robots a beneficial therapeutic tool for children with ASD. Table 2 shows the classification of papers based on target behaviors. In all selected articles, a robot successfully engaged in different activities with the children (e.g. dancing, teaching social skills). These activities are performed in traditional ASD therapy and aim to reduce autistic behaviors and improve the targeted behaviors that CWA struggle with [16]. A robot presented the activities in a game-like format using ABA's reinforcement system in all studies. For example, some papers targeted imitation skills. Imitation is an important way of learning new behaviors [16]. In imitation games, the robot sings, dances, or performs a movement and asks children to repeat the same action. For instance, in Tleubayev et al. [15] children listened to songs and had to imitate dance moves demonstrated by the robot. Joint attention is another social-communicative skill performed in traditional ASD therapy [16]. It refers to the mutual attention focus between two individuals on an object or activity. In Ismail et al. [7] children had to maintain eye contact with the robot, while in Suzuki et al. [14] they had to look at the same object as the robot. CWA have difficulties understanding, recognizing, and expressing emotions [16]. Social robots can teach children to understand emotions through their affective behavior and games [11]. For example, in Sandygulova et al. [11] children learned to recognize and express five emotions. The robot performed the emotions accompanied by a corresponding sound and printed photos to illustrate the emotion and the situation that caused it. Some studies targeted other verbal and non-verbal social gestures such as turn-taking during an activity, greetings, hugging, and handshaking [16]. For example, children in Zhanatkyzy et al. [16] had to wait for their turn to interact with the robot. Another targeted behavior was sensory development. This skill can be improved through sensory stimulation and physical engagement. For example, the robot in Suzuki et al. [14] taught children with autism (CWA) to identify their body parts by touching the robot while dancing. Last but not least, the experiment in Ismail et al. [7] analyzed if stereotyped ASD behavior was reduced by interacting socially with the robot. Stereotyped ASD behavior manifests through repetitive actions, aggression, or self-injury [7]. The only other identified targeted behavior is task completion. Task competition refers to simple tasks that children execute to receive positive feedback from the robot. For example, the robot in Tleubayev et al. [15] asked children to hold its hands.

2.3 Outcomes, Challenges and Gaps

All reviewed articles concluded that the targeted behaviors were improved during the RAAT intervention for all children. Some studies stated that robots could be a valuable tool for traditional ASD therapy since the participants were engaged with the robots e.g. [3, 14]. Table 3 shows the classification of papers based on challenges and gaps and how often they were mentioned, while table 4 classifies the mentioned types of children's heterogeneity. The following subsections are structured based on the identified challenges in the RAAT outcomes.

2.3.1 Heterogeneity of ASD. Some studies compared and analyzed the outcomes of RAAT based on children's heterogeneous subgroups such as LFA vs HFA, or younger vs older children e.g. [11, 16]. All these studies concluded that all targeted behaviors improved for all subgroups thanks to RAAT. However, they also stated that children's differences impacted the outcomes and that few studies in RAAT focus on children's unique profiles [3, 11, 15, 16]. The studies conducted by Sandygulova et al. [11] and Zhanatkyzy et al. [16] grouped children based on: non-verbal LFA vs verbal HFA, the effect of co-occurring ADHD, and age. The study presented in Sandygulova et al. [11] had the following conclusions: verbal children with milder symptoms of ASD (HFA) engaged more with the robot through touch and spoke more words with the robot compared to non-verbal children with LFA. These findings are consistent with the results presented by Zhanatkyzy et al. [16]. In addition, Zhanatkyzy et al. [16] concluded that children with LFA were more curious compared to children with HFA during activities with low-level social mediation (e.g. dancing with the robot) and when the robot did not approach them directly. Regarding the effect of the severity of co-occurring ADHD, Sandygulova et al. [11] concluded that children with severe forms of ASD (LFA) and overlapping ADHD had bigger difficulties expressing emotions through smiling compared to their peers with milder symptoms of ASD and ADHD. The outcomes of the study by Zhanatkyzy et al. [16] are that children with co-occurring ADHD had more challenges keeping their attention during activities that require listening than their peers without ADHD. When it comes to the age factor, both Sandygulova et al. [11] and Zhanatkyzy et al. [16] discovered that older children aged 7-12 y.o. had better behavioral outcomes and more eye contact with the robot compared to younger children aged 5-6 y.o because they interpreted socio-emotional cues better. In addition, younger children aged 3-4 y.o. showed more affective behaviors towards the robot through physical cues such as hugging and touching than their older peers in both studies.

Other studies grouped children by their sensory preferences. The study of Chevalier et al. [3] had two groups: participants with an overreliance on proprioceptive cues and a hyporeactivity to visual cues (group G1) and participants with a hyporeactivity to

Type of Robot	n	Ref. article
NAO	6	[3, 7, 11, 14–16]
humanoid Blue	1	[1]
another humanoid SAR	2	[6, 12]
robotic dog AIBO	1	[13]

Table 1: Types of SAR used in the studies

Targeted Behavior	Ref. article
Imitation	[3, 7, 11, 14–16]
Joint attention	[1, 3, 7, 11, 13–16]
Emotional Skills	[11, 16]
Other Social Skills	[1, 6, 7, 11, 13, 15, 16]
Sensory development	[11, 14–16]
Behavioral reduction	[7, 13]
Task Completion	[12, 15]

Table 2: Targeted Behaviors

proprioceptive cues and an overreliance on visual cues (group G2). Chevalier et al. [3] observed that all children were enthusiastic about participating in the intervention and positively impacted their social skills. However, children from group G1 gazed less toward their robot partner and were less successful in imitating the robot compared to group G2. Chevalier et al. [3] state that robots in RAAT should be able to create sensory profiles for each child before the intervention to personalize activities.

The study by Ismail et al. [7] grouped children based on their IQ following an IQ Test by a certified psychologist. In this study, children with higher IQ levels (above 80) performed better than their peers socially when interacting with the robot and exhibited less stereotyped behaviors during and after RAAT.

Feil-Seifer et al. [6] also outlines the importance of individual differences in CWA on the outcomes. Therefore, Feil-Seifer et al. [6] states that autonomous robots should be improved to interpret a child's behavior. With the help of machine learning (ML), the robot should respond in a timely and appropriate fashion, particularly in instances of negative reactions [6].

2.3.2 HRI Metrics. Begum et al. [1] pointed to the following gap in RAAT research: lack of focus on the efficacy of RAAT because of the commonly used HRI metrics (e.g. gaze towards the robot, verbal/non-verbal communication cues with the robot, affection towards the robot, etc.). Efficacy metrics evaluate how effective a robot is in achieving a targeted therapeutic effect [1]. However, the common HRI metrics could not deliver enough information about the efficacy of the robot in achieving the therapy objective [1]. Consequently, Begum et al. [1] indicate the need to define better efficacy metrics for RAAT that consider children's differences.

2.3.3 Long-term outcomes in real-world deployment . Another gap in the RAAT research identified by the reviewed papers (e.g. [3, 11, 12, 15, 16]) consists in the lack of studies about long-term outcomes of RAAT in real-world scenarios.

Challenges and Gaps	n	Ref. article
Children's heterogeneity	8	[1, 3, 6, 7, 11, 12, 15, 16]
HRI Metrics	1	[1]
Long-term in real world	4	[1, 11, 12, 16]

Table 3: Challenges and gaps in RAAT

Children's heterogeneity	Ref. article
Verbality HFA vs LFA	[11, 16]
ADHD	[11, 15, 16]
Sensory preferences	[3, 6]
Age	[11, 16]
Cognitive level (IQ)	[7]
Automated response	[6, 12]

Table 4: Challenges and gaps in RAAT: Children's heterogeneity

3 METHODOLOGY

This systematic literature review was conducted following the PRISMA guidelines, and utilizing the ACM digital library for research. Fig. 1 shows the PRISMA flow diagram.

After gaining a general understanding of the topic, I concluded that one of the primary uses of robots in mental health is for therapy for children with autism. Therefore, I decided to investigate this topic because of its relevance. After trying different keyword combinations, I selected the search query: "hri AND autism AND robot AND therapy" with the filter "Research Article". The applied search query led to 122 results.

I manually screened the 122 results to obtain the most suitable literature for my purpose. Firstly, the articles were filtered by their title. Consequently, 83 articles were excluded. In the second round, I read the abstract section of the remaining 39 papers. I excluded 18 articles because they didn't focus on RAAT. In the final round, I read the full text of the remaining 21 articles and excluded articles that didn't focus on my research questions. I referenced 16 articles in my systematic review. I included the studies from 10 articles and I excluded 6 articles because they involved participants other than children, lacked a specific focus on RAAT, or focused on technical details.

4 DISCUSSION

The findings suggest that RAAT can be a valuable addition to ASD therapy. Robots are meant to act as assistants between children and the therapist and not to replace human therapists [11]. Firstly, the used robots could perform activities specific to traditional ASD therapy based on ABA principles with no difficulties in all reviewed studies. Secondly, all studies concluded that the targeted behaviors were improved after the RAAT session for all participants. Suzuki et al. [14] noticed no major differences in completing the tasks between the human and the robot-led therapy. Moreover, robots are meant to make therapists' jobs easier. CWA tend to learn better through repetition, which a robot can provide without getting tired, unlike a human therapist [15]. Furthermore, most CWA are engaged and receptive when learning with robots [11, 14–16].

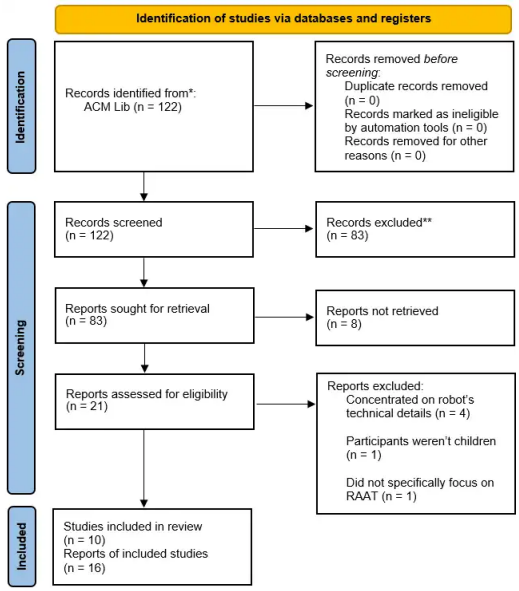


Figure 1: Figure 1: PRISMA flow diagram

Therefore, robots could be effective therapy tools, motivating CWA to participate in activities [5, 11, 15, 16].

On the other hand, working with robots can bring a set of challenges. One of the most mentioned challenges in RAAT is the lack of studies on children's heterogeneity [11, 16]. The most used robot in the reviewed studies is the autonomous social humanoid NAO. NAO has been the primarily used robot for RAAT in the past decade [11]. However, NAO has its behavioral and response limitations. While machine learning (ML) training could be challenging because of costs or time, the main hurdle is training ML for children's unique differences. The findings from the reviewed studies demonstrate that some RAAT activities are more effective for children with HFA and less autistic behaviors [11, 16]. Children who are more to the lower end of the ASD spectrum benefited less from the standard RAAT interventions [16]. Moreover, sensory preferences are also important when designing the robot according to Chevalier et al. [3]. That being said, there cannot be a general-purpose robot that would benefit all children equally since children's diversity influences the intervention outcomes[11]. Therefore, the RAAT activities have to be customized based on children's unique profiles. However, a more complex robot that adapts to children's differences could be challenging to implement and requires time and money for testing.

Furthermore, the development and testing of a robot that can automatically react to children's responses in real-time can be challenging. Robots like NAO are not fully autonomous and have their response limitations. However, Feil-Seifer et al. [6] highlights the importance of automated detection and its influence on the intervention outcomes.

Last but not least, HRI metrics for measuring the efficacy of RAAT is another identified gap that needs further research according to Begum et al. [1].

Another important identified gap in RAAT research is the lack of studies in real-world scenarios for long-term interventions [11, 16]. The reviewed studies were conducted within controlled settings and evaluated short-term outcomes. On the one hand, studies on long-term interventions or in real-world scenarios can be challenging because of costs, time, or maintaining participants over the long-term period. On the other hand, research on long-term outcomes in real-world scenarios is necessary for the well-being of children with ASD. Despite these limitations, further research is needed to compare the long-term outcomes of groups of children who received only traditional ASD therapy and those who also received robot-assisted therapy. ASD therapy is meant to be effective not only in the controlled therapeutic environment but also in the real world over longer periods.

Despite all the previously mentioned barriers, RAAT has great potential to not only improve children's experiences in therapy sessions but also help CWA in the long term in their own lives. There exists no universal solution, therapy, or robot that can equally benefit all children with Autism Spectrum Disorder (ASD). Although HRI may not achieve perfection, it can get closer to implementing robot-assisted therapy interventions that are better aware of the heterogeneous form of autism and the importance of automated detection of children's reactions. More in-depth research on the aspect of heterogeneity could aid in improving the robot's range of behaviors, reactions, and activities. The robot should easily identify and respond to negative reactions from children.

5 CONCLUSION

Based on the findings of this systematic literature review, socially assistive robots have great potential to improve traditional ASD therapy, and as a result, the lives of children with ASD. The humanoid robots in the reviewed studies could successfully perform activities specific to traditional ASD therapy and improve targeted behaviors (e.g. imitation, social skills, joint attention, emotional skills, etc.). SARs could make therapists' jobs easier since they don't get tired from repeating activities. Additionally, research shows that Children with autism (CWA) are engaged with robots. Consequently, RAAT could improve their motivation and attention during the sessions.

Nevertheless, RAAT research faces some challenges and research gaps such as children's heterogeneity and lack of research on long-term studies in real-world settings. Children with ASD differences (e.g. level of verbality, co-occurrence of ADHD, etc.) influence the intervention outcomes. However, the currently most used SAR in RAAT, Nao, is limited in its behavior and cannot properly adjust to children's unique profiles. Another identified gap in RAAT research is the common HRI metrics that can not deliver enough information about the efficacy of RAAT.

This systematic review flags a need for more RAAT studies on children's heterogeneity. RAAT research should prioritize the identification and classification of subgroups among CWA based on their heterogeneity. Subsequently, robots' capabilities could be adjusted to children's subgroups and respond to children's positive or negative reactions in an appropriate manner. This way, CWA from all subgroups could benefit more from RAAT interventions. Furthermore, RAAT research should improve robots' capabilities, improve current HRI metrics, and conduct more long-term studies on the efficacy of RAAT in real-world settings.

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